

# Overview of Prompt engineering

*Subject: Prompt engineering*

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## 1.

**Limitations** The process of writing and refining a prompt for an LLM or generative AI shares some parallels with an iterative engineering design process, such as by discovering reusable best practices through reproducible experimentation. But the techniques that improve performance depend heavily on the specific model being used. Such patterns are also volatile and exhibit significantly different results from seemingly insignificant prompt changes.

## 2.

**Using language models to generate prompts** LLMs themselves can be used to compose prompts for LLMs. The automatic prompt engineer algorithm uses one LLM to beam search over prompts for another LLM:

## 3.

**MIPRO** (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. **GEPA** (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

## 4.

**Multi-shot** A prompt may include a few examples for a model to learn from in context, an approach called few-shot learning. For example, the prompt may ask the model to complete "maison → house, chat → cat, chien →", with the expected response being dog.

## 5.

**Prompt engineering** is the process of structuring natural language inputs (known as prompts) to produce specified outputs from a generative artificial intelligence (GenAI) model. **Context engineering** is the related area of software engineering that focuses on the management of non-prompt contexts supplied to the GenAI model, such as metadata, API tools, and tokens. It can also be defined as the practice of designing and refining input instructions given to a generative AI model to produce more accurate, relevant, or useful outputs. Effective prompt engineering involves understanding how a model interprets language, and may include techniques such as few-shot prompting, chain-of-thought prompting, and role assignment. It is increasingly considered a skill for working with large language models (LLMs) in both research and professional contexts. During the 2020s AI boom, prompt

engineering became regarded as a business capability across corporations and industries. Employees with the title prompt engineer were hired to create prompts that would increase productivity and efficacy, although the individual title has since lost traction amid AI models that produce better prompts than humans and corporate training in prompting for general employees. Common prompting techniques include multi-shot, chain-of-thought, and tree-of-thought prompting, as well as the use of assigning roles to the model. Automated prompt generation methods, such as retrieval-augmented generation (RAG), provide for greater accuracy and a wider scope of functions for prompt engineers. Prompt injection is a type of cybersecurity attack that targets machine learning models through malicious prompts.

## **6.**

GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights, and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

## **7.**

Techniques Common terms used to describe various specific prompt engineering techniques include chain-of-thought, tree-of-thought, and retrieval-augmented generation (RAG). A 2024 survey of the field identified over 50 distinct text-based prompting techniques, 40 multimodal variants, and a vocabulary of 33 terms used across prompting research, highlighting a present lack of standardised terminology for prompt engineering. Vibe coding is an AI-assisted software development method where a user prompts an LLM with a description of what they want and lets it generate or edit the code. In 2025, "vibe coding" was the Collins Dictionary word of the year.

## **8.**

In 2022, text-to-image models like DALL-E 2, Stable Diffusion, and Midjourney were released to the public. These models take text prompts as input and use them to generate images. Early text-to-image models typically do not understand negation, grammar and sentence structure in the same way as large language models, and may thus require a different set of prompting techniques. The prompt "a party with no cake" may produce an image including a cake.

## **9.**

Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

## **10.**

Textual inversion and embeddings For text-to-image models, textual inversion performs an optimization process to create a new word embedding based on a set of example images. This embedding vector acts as a "pseudo-word" which can be included in a prompt to express the content or style of the examples.

## **11.**

**Image prompting** In 2023, Meta's AI research released Segment Anything, a computer vision model that can perform image segmentation by prompting. As an alternative to text prompts, Segment Anything can accept bounding boxes, segmentation masks, and foreground/background points.

## **12.**

**Context engineering** Context engineering is a related process that focuses on the context elements that accompany user prompts, which include system instructions, retrieved knowledge, tool definitions, conversation summaries, and task metadata. Context engineering is performed to improve reliability, provenance and token efficiency in production LLM systems. The concept emphasises operational practices such as token budgeting, provenance tags, versioning of context artifacts, observability (logging which context was supplied), and context regression tests to ensure that changes to supplied context do not silently alter system behaviour.

## **13.**

A text-to-image prompt commonly includes a description of the subject of the art, the desired medium (such as digital painting or photography), style (such as hyperrealistic or pop-art), lighting (such as rim lighting or crepuscular rays), color, and texture. Word order also affects the output of a text-to-image prompt. Words closer to the start of a prompt may be emphasized more heavily.